

From Self-excitation to Saturation: The Riga Dynamo Story

Sampreet Das
2025UG029
AEC 2101 (Instructor: Prof. Raja Paul)

Overview: This investigative study covers one of the first laboratory verification of homogeneous hydromagnetic self-excitation in liquid sodium. We begin with some pre-requisite mathematical formulations and theory, including Ponomarenko's screw model. Then we study two seperate regimes, namely- kinematic and saturation regime to explain self-excitation and saturation respectively, and draw conclusions about how this dynamo setup works. Intuitive understanding is prioritised over rigorous mathematics in this study.

Two Formulae and some theoretical understanding

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \lambda \nabla^2 \mathbf{B} \longrightarrow \mathbf{B}(t) \propto e^{(p+i2\pi f)t}$$

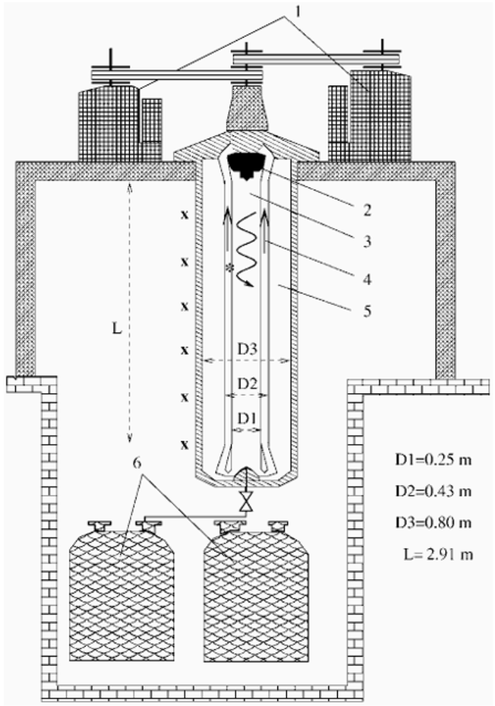
$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{1}{\mu_0 \rho}(\nabla \times \mathbf{B}) \times \mathbf{B} + \nu \nabla^2 \mathbf{v} + \mathbf{f}_{\text{drive}} \qquad Rm = \mu_0 \sigma L v$$

- A **hydromagnetic dynamo** is a self-sustaining model which undergoes **self-excitation** of magnetic field due to a certain type of motion of an electrically conducting fluid, which gives rise to a non axis-symmetrical **rotating magnetic field**.

- Upon reaching a **critical threshold** (R_{mc}), it triggers saturation, which prevents its exponential growth to infinity by applying a Lorentz force to change the local velocity of the fluid, vanishing the growth rate.

- **Ponomarenko's helical screw model**: The Riga dynamo experiment is the technical realization of what could be called an "elementary cell" of all hydromagnetic dynamos: a screw motion. The main idea traces back to the paper of Ponomarenko (1973) who considered magnetic field self-excitation caused by the screw motion of an electrically **conducting solid body** within a medium, at rest, with the same conductivity.

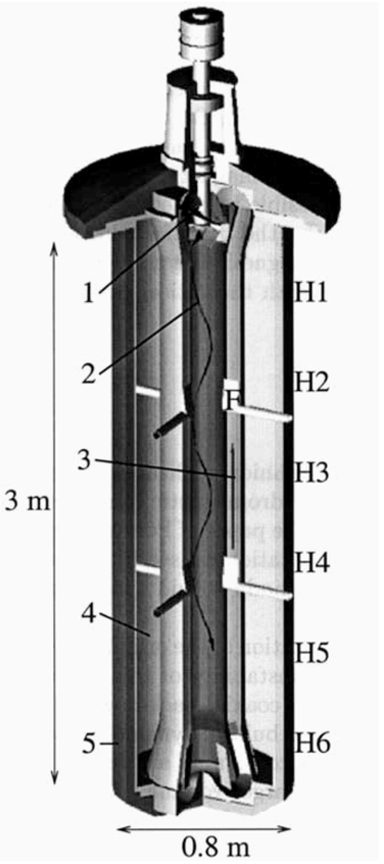
Central module design: optimising to achieve self-excitation



Fluid used: Liquid Sodium

Dynamo Facility Architecture: 3 Co-Axial cylinders

- **Inner Tube**: Contains primary liquid sodium helical screw flow, driven directly by a top-mounted propeller (approx. power 120kW, max. Rm achievable).
- **Middle Channel**: Directs sodium in the back-flow direction to convert convective instability into abosulte instability.
- **Outer Shell**: Contains Na (liq.) at rest for reducing Ohmic dissipation losses



Kinematic theory shows that the resulting self-excited magnetic field continuously **rotates** around the vertical axis in the **same direction as the fluid rotation**, but at a much **slower speed**.

Dealing with the Velocity Profile:

Issue: Azimuthal flow concentrated heavily near the central axis, dropping off toward the outer edge.
Consequence: Magnetic Renoylds number = 31 (far exceeding the 120 kW power limit)
Solution: Custom vane arrays were installed immediately in front and behind the propeller

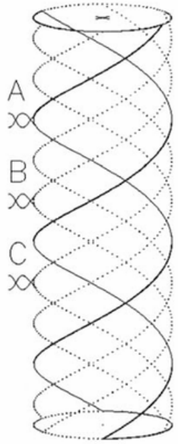
Tailored velocities:

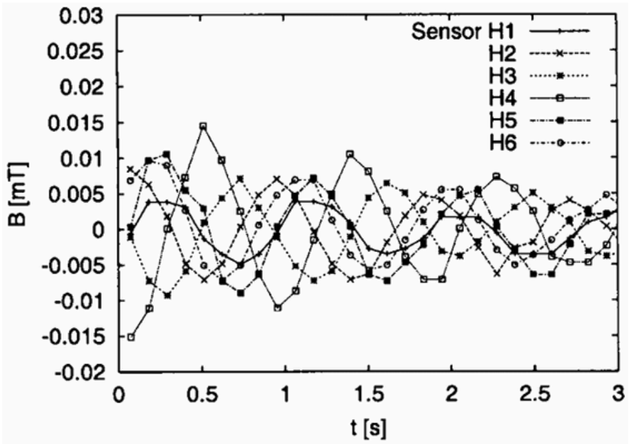
At approx. 2000 rpm, axial velocity 15 m/s, azimuthal velocity 9m/s
(Detected by Laser-Doppler-Velocimetry (LDV))
Turbulent velocity fluctuations were measured at less than 10% of the mean flow speed, hence negligible

Phase 1: Achieving Self-Excitation (November 1999)

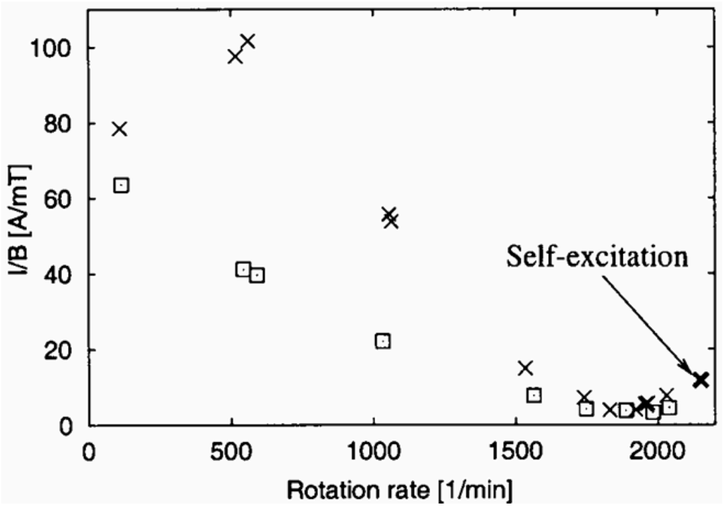
Liq. Na pumped at 300 $^{\circ}C$ for 24 hrs

Seed field: External 3-phase seed field coils wrapped around the central module produced a 3D rotating seed field matching the expected spatial structure of the dynamo eigenmode (feeding frequency 1 Hz)





Part A: Subcritical regime
At propeller rotation rates **below the critical threshold** (e.g., 1980 rpm), switching off the current in the seed coils caused the magnetic field signal to **decay** (at 1980 rpm, p = -0.3 s-1 and f = 1.1 Hz).

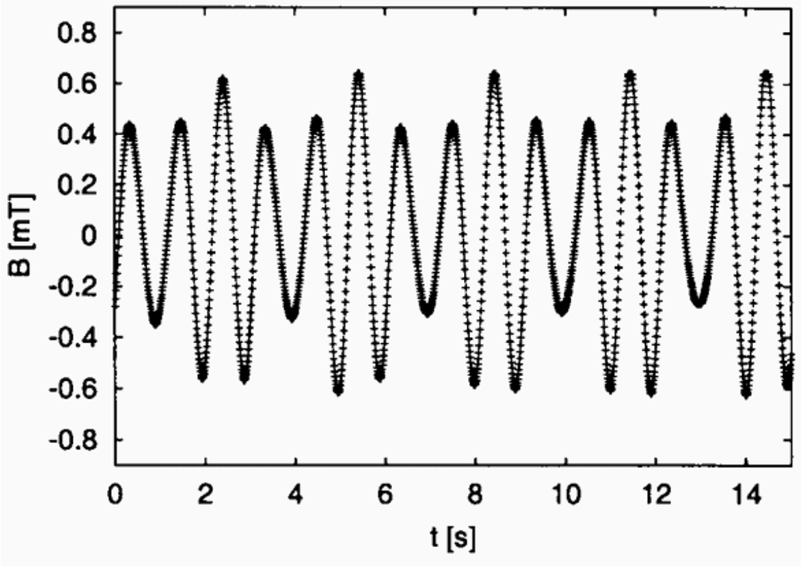


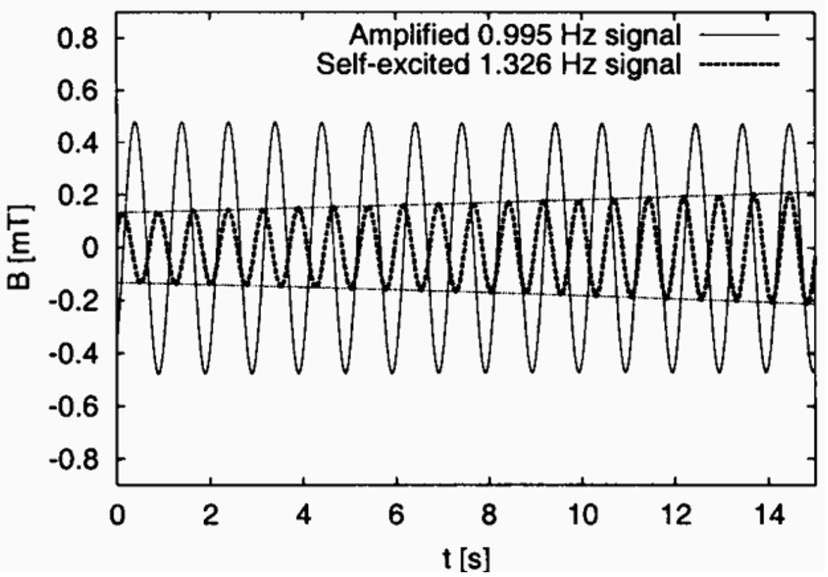
Part B: Supercritical regime

Signal decomposition revealed **two independent modes** operating simultaneously:

- An amplified driving mode matching the **seed field frequency** (0.995 Hz).
- A **self-excited mode** characterized by a frequency of f = 1.326 Hz and a positive growth rate of p = +0.03 s^{-1}.

As predicted by kinematic dynamo theory, after the acceleration phase, all non-eigenmode frequencies decayed, leaving **only the natural eigenmode** of the dynamo to grow exponentially over time.





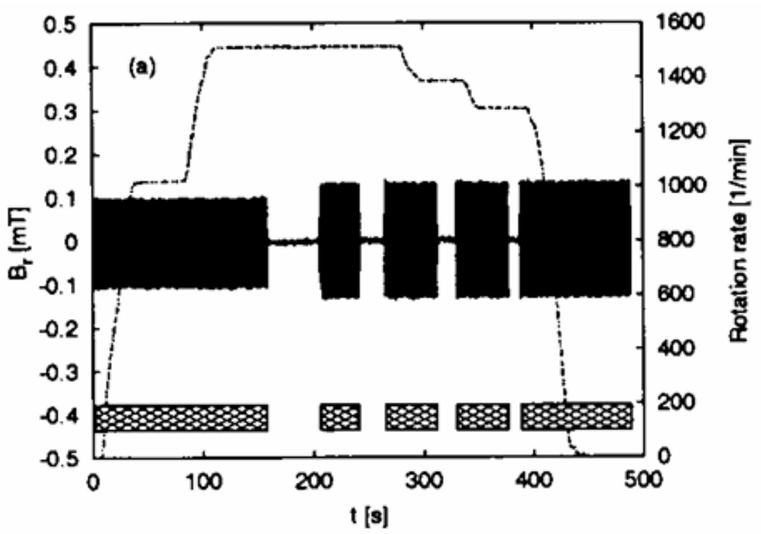
Phase 2: Achieving Saturation (July 2000)

Lower temperatures used mostly (around 160 $^{\circ}C$)

Four runs conducted at different temperatures and seed fields:

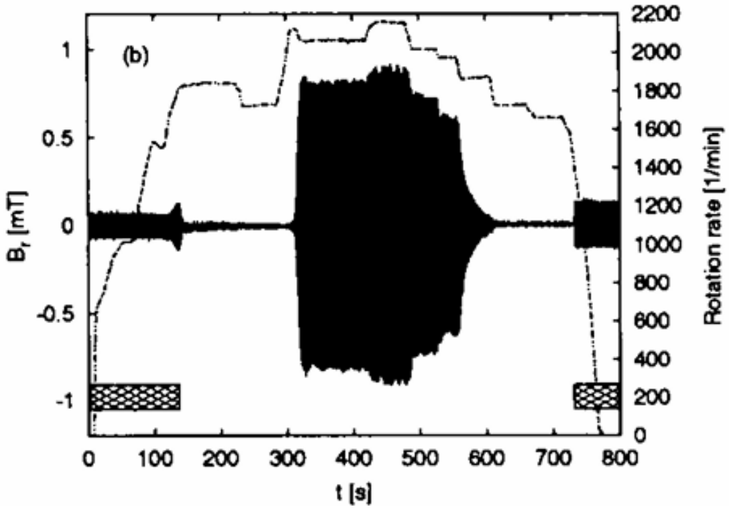
Run 1:

- Seed field frequency = 1 Hz, Temperature= 210 $^{\circ}C$
- Conducted entirely in the subcritical regime; recorded only the amplified 1 Hz seed field signal



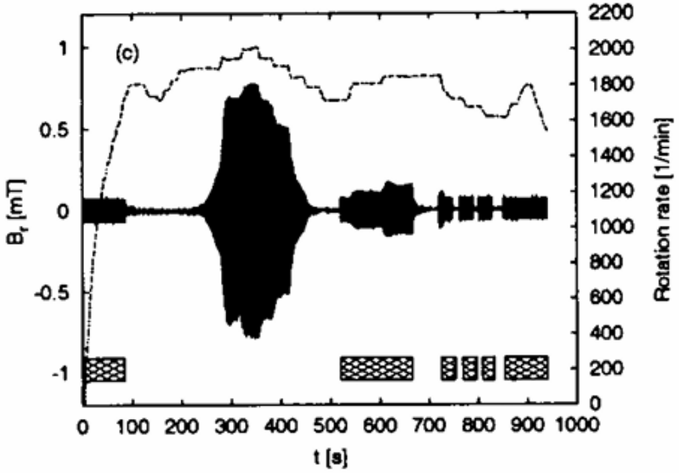
Run 2:

Seed field frequency = 1 Hz, Temperature= 160$^{\circ}C$

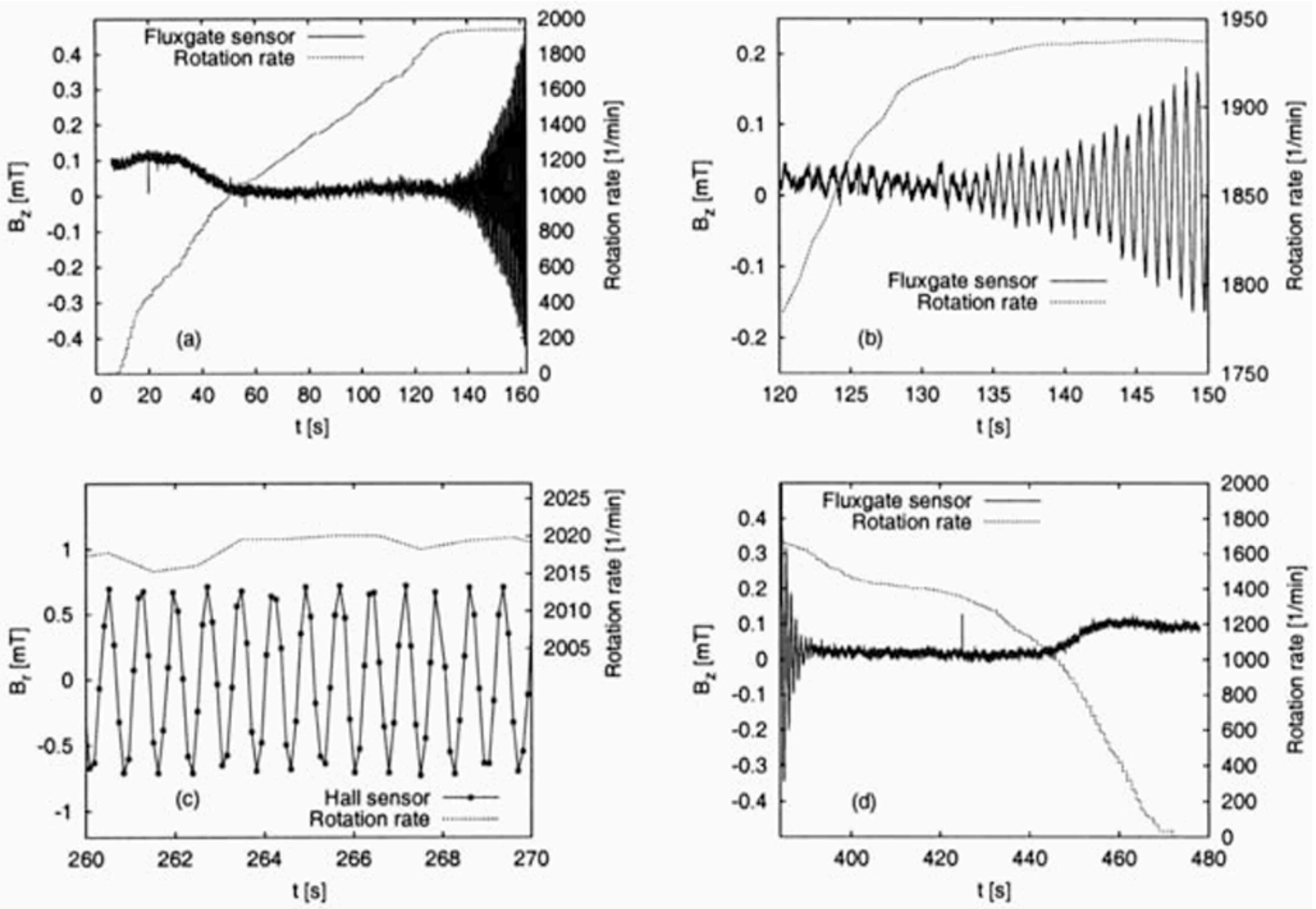


Run 3:

Intermittent seed field, Temperature= 160 $^{\circ}C$



Run 4: No Seed field, Temperature= 160 $^{\circ}C$



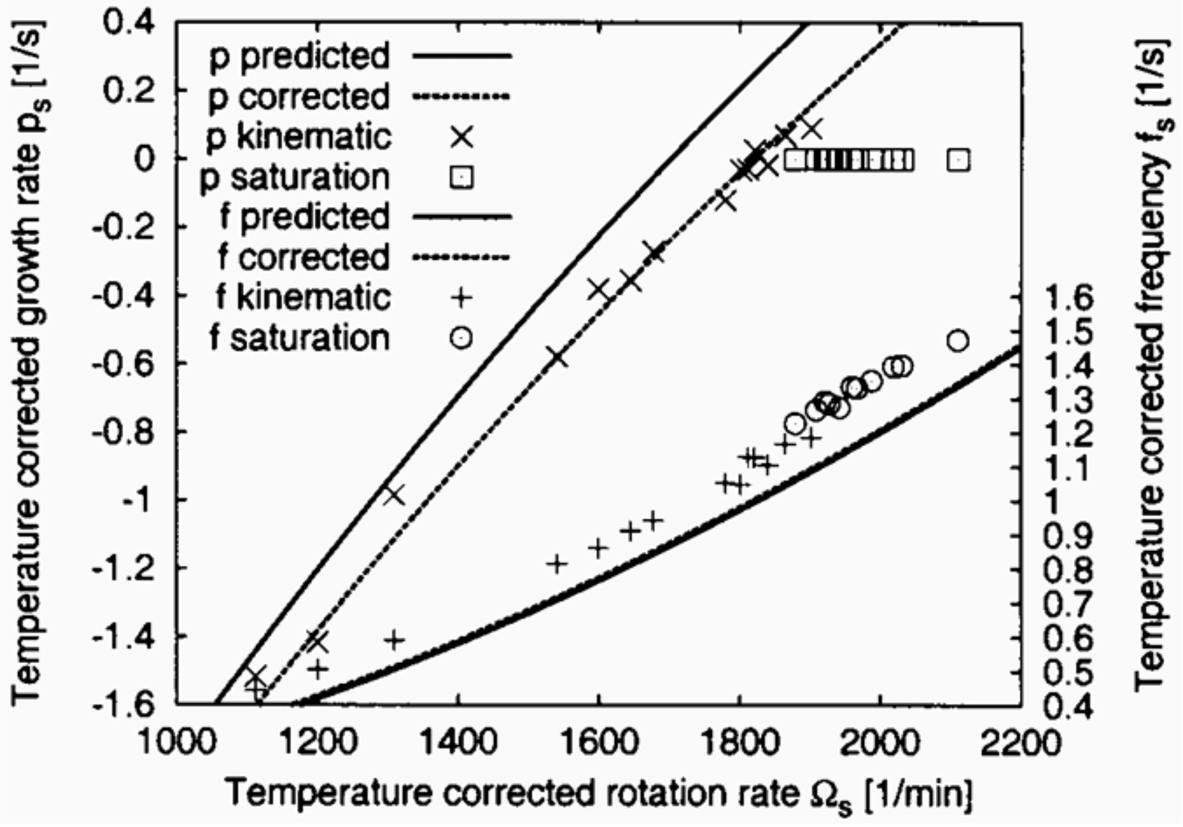
Net growth rate **dropping to zero** (p = 0).

The self-excited magnetic signal spontaneously "emerged from background noise" once the rotation rate crossed the critical threshold (1920 rpm) and saturated at steady levels corresponding to the motor speed.

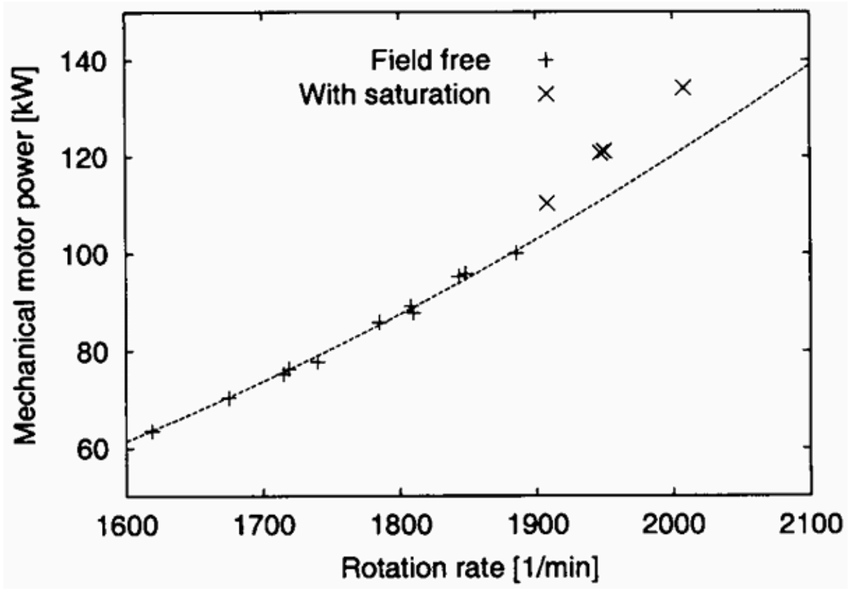
Interpreting the major experimental data

Temperature-corrected scaling:

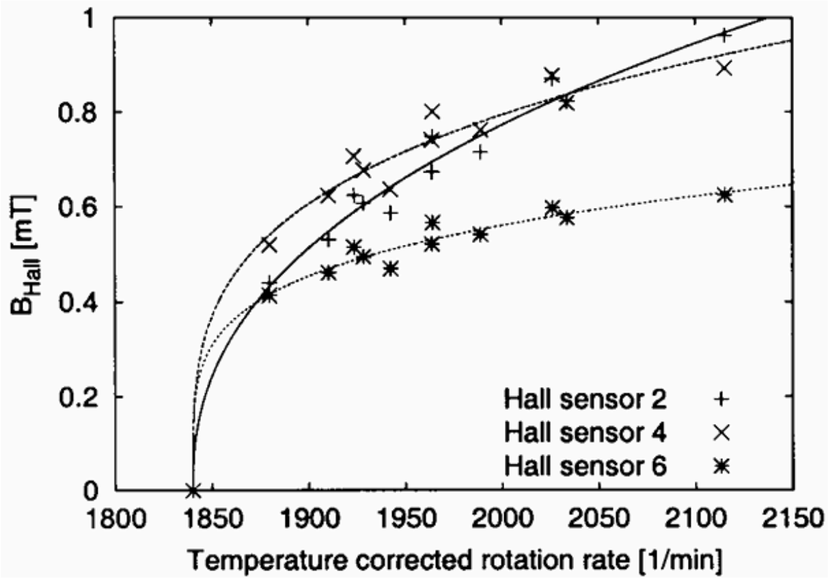
$$(\Omega_s,\, p_s,\, f_s) = \sigma(T)/\sigma(157\,^{\circ}\mathrm{C})(\Omega(T),\, p(T)\, f(T))$$



Trend of growth rate and frequencies over all regime



Power consumption vs. Rotation rate

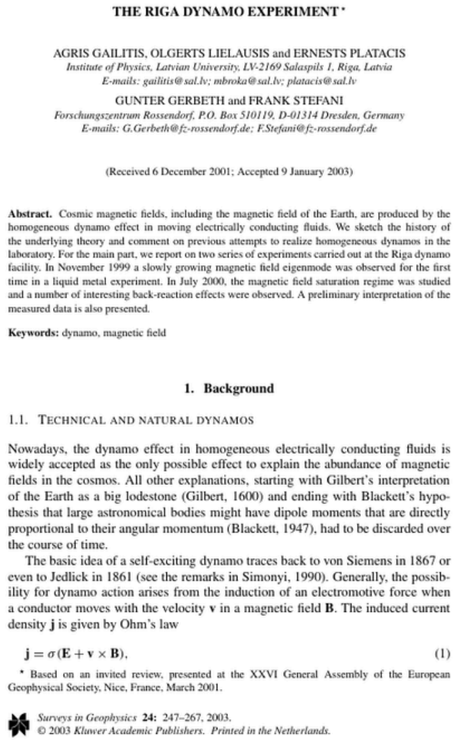


Evidence of magnetic field shifting upwards

Possible future prospects

- **The Karlsruhe Dynamo Experiment**: While Riga realized a single-scale, localized helical screw dynamo (Ponomarenko model) producing an oscillating, rotating magnetic field, the Karlsruhe facility demonstrated a homogeneous two-scale alpha square-dynamo. Using 52 spin-generator channels (helically linked sodium ducts), it showed how small-scale helical fluid motions organize to produce a large-scale, stationary magnetic dipole.
- **Solar Flux-Transport Dynamos & Magnetic Cycles**: Explore how the combined action of differential rotation (Omega-effect) and meridional circulation creates the 11-year/22-year solar cycle and sunspot butterfly diagrams, connecting the periodic phase behavior of Riga's magnetic eigenmodes to stellar magnetic cycles
- **Creating a feedback loop simulation**: As we discussed, the Navier-Stokes momentum equation and the induction equation works together to obtain a saturation regime in the moving fluid. Now this dependancy can possibly be simulated to generate magnetic fields and that simulation can be probed to obtain a possibly cost-effective way to perform some propective experiments.

Citations and References



THE RIGA DYNAMO EXPERIMENT*

AGRIS GAILITIS, OLGERTS LIELAUSIS and ERNESTS PLATACIS
Institute of Physics, Latvian University, LV-2169 Salaspils 1, Riga, Latvia
E-mails: gailitis@sal.lv; mbroka@sal.lv; platacis@sal.lv
GUNTER GERBETH and FRANK STEFANI
Forschungszentrum Rossendorf, P.O. Box 510119, D-01314 Dresden, Germany
E-mails: G.Gerbeth@fz-rossendorf.de; F.Stefani@fz-rossendorf.de

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Abstract. Cosmic magnetic fields, including the magnetic field of the Earth, are produced by the homogeneous dynamo effect in moving electrically conducting fluids. We sketch the history of the underlying theory and comment on previous attempts to realize homogeneous dynamos in the laboratory. For the main part, we report on two series of experiments carried out at the Riga dynamo facility. In November 1999 a slowly growing magnetic field eigenmode was observed for the first time in a liquid metal experiment. In July 2000, the magnetic field saturation regime was studied and a number of interesting back-reaction effects were observed. A preliminary interpretation of the measured data is also presented.

Keywords: dynamo, magnetic field

1. Background

1.1. Technical and Natural Dynamos

Nowadays, the dynamo effect in homogeneous electrically conducting fluids is widely accepted as the only possible effect to explain the abundance of magnetic fields in the cosmos. All other explanations, starting with Gilbert's interpretation of the Earth as a big lodestone (Gilbert, 1600) and ending with Blackett's hypothesis that large astronomical bodies might have dipole moments that are directly proportional to their angular momentum (Blackett, 1947), had to be discarded over the course of time.

The basic idea of a self-exciting dynamo traces back to von Siemens in 1867 or even to Jedlick in 1861 (see the remarks in Simonyi, 1990). Generally, the possibility for dynamo action arises from the induction of an electromotive force when a conductor moves with the velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$. The induced current density $\mathbf{j}$ is given by Ohm's law

$$\mathbf{j} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \tag{1}$$

* Based on an invited review, presented at the XXVI General Assembly of the European Geophysical Society, Nice, France, March 2001.

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The Riga Dynamo Experiment

Gailitis, A., Lielausis, O., Platacis, E. et al.

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Detection of a Flow Induced Magnetic Field Eigenmode in the Riga Dynamo Facility

Agris Gailitis, Olgerts Lielausis, Sergej Dement'ev, Ernests Platacis, and Arnis Cifersons
Institute of Physics, Latvian University, LV-2169 Salaspils 1, Riga, Latvia
Gunter Gerbeth, Thomas Gundrum, and Frank Stefani
Forschungszentrum Rossendorf, P.O. Box 510119, D-01314 Dresden, Germany
Michael Christen, Heiko Hänel, and Gotthard Will
Department of Mechanical Engineering, Dresden University of Technology, P.O. Box 01062, Dresden, Germany
(Received 10 December 1999)

In a closed volume of molten sodium an intense single-vortex-like helical flow has been produced by an outside powered propeller. At a flow rate of 0.67 m^3/s a slowly growing magnetic field eigenmode was detected. For a slightly lower flow, additional measurements showed a slow decay of this mode. The measured results correspond satisfactorily with numerical predictions for the growth rates and frequencies.

PACS numbers: 47.65.+a, 52.65.Kj, 91.25.Cw

Magnetic fields of cosmic bodies, such as the Earth, most of the planets, stars, and even galaxies, are believed to be generated by the dynamo effect in moving electrically conducting fluids. Whereas technical dynamos consist of a number of well-separated electrically conducting parts, a cosmic dynamo operates, without any ferromagnetism, in a nearly homogeneous medium (for an overview, see, e.g., [1] and [2]).

The governing equation for the magnetic field $\mathbf{B}$ in an electrically conducting fluid with conductivity σ and the velocity $\mathbf{v}$ is the so-called induction equation

$$\frac{\partial \mathbf{B}}{\partial t} = \mathrm{curl}(\mathbf{v} \times \mathbf{B}) + \frac{1}{\mu_0 \sigma} \Delta \mathbf{B}, \tag{1}$$

which follows from Maxwell equations and Ohm's law. The obvious solution $\mathbf{B} = 0$ of this equation may become unstable for some critical value Rm_c of the *magnetic Reynolds* number $\mathrm{Rm} = \mu_0 \sigma L v$ if the velocity field fulfills some additional conditions. Here L is a typical length scale, and v a typical velocity scale of the fluid system. Rm_c depends strongly on the flow topology and the helicity of the velocity field. For self-excitation of a magnetic field it has to be at least greater than 1. For typical dynamos as the Earth outer core, Rm is supposed to be of the order of 100.

The last decades have seen enormous progress in dynamo theory which deals, in its kinematic version, with the induction equation exclusively or, in its full version, with the coupled system of induction equation and Navier-Stokes equation for the fluid motion. Numerically, this coupled system of equations has been treated for a number of more or less realistic models of cosmic bodies (for an impressive simulation, see [3]).

Quite contrary to the success of dynamo theory, experimental dynamo science is still in its infancy. This is mainly due to the large dimensions of the length scale and/or the velocity scale which are necessary for dynamo action to occur. Considering the conductivity of sodium as one of the best liquid conductors [$\sigma \approx 10^7$ $(\Omega\,\mathrm{m})^{-1}$ at 100 °C] one gets $\mu_0 \sigma \approx 10\ \mathrm{s/m^2}$. For a very efficient dynamo with a supposed $\mathrm{Rm}_c = 10$ this would amount to a necessary product $Lv = 1\ \mathrm{m^2/s}$, which is very large for a laboratory sodium facility if one takes into account the safety problems. Historically notable for experimental dynamo science is the experiment of Lowes and Wilkinson where two ferromagnetic metallic rods were rotated in a block at rest [4]. A first liquid metal dynamo experiment quite similar to the present one was undertaken by some of the authors in 1987. Although this experiment had to be stopped (for reasons of mechanical stability) before dynamo action occurred the extrapolation of the amplification factor of an applied magnetic field gave indication for the possibility of magnetic field self-excitation at higher pump rates [5]. Today, there are several groups working on liquid metal dynamo experiments. For a summary we refer to the workshop "Laboratory Experiments on Dynamo Action" held in Riga in the summer of 1998 [6].

After years of preparation and careful velocity profile optimization on water models, first experiments at the Riga sodium facility were carried out during 6–11 November 1999. The present paper comprises only the most important results of these experiments, one of them being the observation of a dynamo eigenmode slowly growing in time at the maximum rotation rate of the propeller. A more comprehensive analysis of all measured data will be published elsewhere.

The principal design of the dynamo facility, together with some of the most important dimensions, is shown in Fig. 1. The main part of the facility consists of a spiral flow of liquid sodium in an innermost tube (with a velocity up to the order of 15 m/s) with a coaxial backflow region and a

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Detection of a Flow Induced Magnetic Field Eigenmode in the Riga Dynamo Facility

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Magnetic Field Saturation in the Riga Dynamo Experiment

Agris Gailitis, Olgerts Lielausis, Ernests Platacis, Sergej Dement'ev, and Arnis Cifersons
Institute of Physics, Latvian University, LV-2169 Salaspils 1, Riga, Latvia
Gunter Gerbeth, Thomas Gundrum, and Frank Stefani
Forschungszentrum Rossendorf, P.O. Box 510119, D-01314 Dresden, Germany
Michael Christen and Gotthard Will
Department of Mechanical Engineering, Dresden University of Technology, P.O. Box 01062, Dresden, Germany
(Received 18 October 2000)

After the dynamo experiment in November 1999 [A. Gailitis *et al.*, Phys. Rev. Lett. **84**, 4365 (2000)] had shown magnetic field self-excitation in a spiraling liquid metal flow, in a second series of experiments emphasis was placed on the magnetic field saturation regime as the next principal step in the dynamo process. The dependence of the strength of the magnetic field on the rotation rate is studied. Various features of the saturated magnetic field are outlined and possible saturation mechanisms are discussed.

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In the last few decades, the theory of homogeneous dynamos has been unrivaled in explaining magnetic fields of planets, stars, and galaxies. Enormous progress has been made in the numerical treatment of the so-called kinematic dynamo problem where the velocity $\mathbf{v}$ of the fluid with electrical conductivity σ is assumed to be given and the behavior of the magnetic field $\mathbf{B}$ is governed by the induction equation

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \frac{1}{\mu_0 \sigma} \Delta \mathbf{B}. \tag{1}$$

Above a critical value of the magnetic Reynolds number $Rm = \mu_0 \sigma L v$, where L and v denote a typical length and velocity scale of the fluid, respectively, the obvious solution $\mathbf{B} = 0$ of Eq. (1) may become unstable and self-excitation of a magnetic field may occur. The investigation of the saturation mechanism which limits the exponential growth of the magnetic field is much more involved as it requires the simultaneous solution of Eq. (1) and the Navier-Stokes equation

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{1}{\mu_0 \rho} (\nabla \times \mathbf{B}) \times \mathbf{B} + \nu \Delta \mathbf{v}, \tag{2}$$

where the back-reaction of the magnetic field on the velocity is included. In Eq. (2), ρ and ν denote the density and the kinematic viscosity of the fluid, respectively.

A number of recent computer simulations of the Earth's core have led to impressive similarities between the computed and the observed magnetic field behavior including, most remarkably, field reversals [1]. The actual relevance of these simulations for real dynamos is, however, not completely clear as they are either working in parameter regions far from that of the Earth or have to resort to numerical trickery as, e.g., the use of anisotropic hyperdiffusivities (for a recent overview, see [2]).

Until very recently, a serious problem of the science of hydromagnetic dynamos was the lack of any possibility to verify numerical results by experimental work. This situation changed on 11 November 1999 when in the first Riga dynamo experiment self-excitation of a slowly growing magnetic field eigenmode was observed for the first time in an experimental liquid metal facility [3]. On the background of an amplified external signal, the growth rate and the rotation frequency of the self-excited field were identified to be in good agreement with the numerical predictions based on kinematic dynamo theory. Because of some technical problems, the saturation regime was not reached in this first experiment. Soon after, Müller and Stieglitz studied self-excitation and the saturation regime in another dynamo facility in Karlsruhe [4].

In the following we represent the results of a second series of experiments at the Riga facility which were carried out during 22–25 July 2000. It was possible to work at considerably lower sodium temperatures and thus at higher electrical conductivities than in the November experiment. This fact allowed us to study in great detail the dynamo behavior in the kinematic as well as in the saturation regime. Here, we mainly focus on the results in the saturation regime as they go essentially beyond the results of the November 1999 experiment.

The Riga dynamo facility consists of three concentric tubes of approximately 3 m length (Fig. 1). In the innermost tube of 25 cm diameter a spiraling flow of liquid sodium with a velocity up to 15 m/s is produced by a propeller. The sodium flows straight back in a second coaxial tube and stays at rest in a third outermost tube. For more details of the experimental design, see [3] and references therein.

As in November 1999, the r component of the magnetic field outside the dynamo was measured by Hall sensors at eight different places, six of which were aligned parallel to the axis. Within the dynamo, close to the

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Magnetic Field Saturation in the Riga Dynamo Experiment

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Phys.Rev.Lett.86, 3024–3027

Special Mention: Ponomarenko, Yu. B.: 1973, 'On the Theory of Hydromagnetic Dynamo', J. Appl. Mech. Tech. Phys. 14, 775–779

Thank You